

Research Topic

A Review on AI/ML-Driven Human Milk Banking: Infrastructure, Social Barriers and Future Directions.

Abstract

Human milk is recognized as the optimal source of nutrition for infants, particularly for premature and low-birth-weight babies. However, donation rates and milk bank efficiency remain limited in many regions due to social, cultural, and technical barriers. This review consolidates findings from nine recent studies covering awareness, donor motivation, quality improvement, and emerging applications of artificial intelligence (AI) and machine learning (ML) in human milk banking. The results reveal that behavioral and educational factors strongly influence donation willingness, while process-level interventions enhance milk collection. Concurrently, ML and AI technologies show great potential for nutrient prediction, digital monitoring, and process automation. Integrating human-centered strategies with AI-driven innovation can transform milk banking into a more efficient, transparent, and sustainable ecosystem.

Short Background

Human Milk Banking (HMB) plays a vital role in ensuring access to safe donor milk for infants when a mother's milk is unavailable. Despite its significance, challenges such as low awareness, social myths, and limited infrastructure restrict milk bank operations. Recent studies explore behavioral, institutional, and technological perspectives, revealing that a combination of educational initiatives and digital tools is essential for sustainable HMB growth.

Research Paper Titles

1. Ajinkya

Paper 1: Predicting Protein and Fat Content in Human Donor Milk Using Machine Learning

Author: Raghavan A. K. et al. (2021)

Abstract Summary: Applied ML models such as OLS, Lasso, Random Forest, and Gradient Boosting to predict protein and fat levels in donor milk. Achieved high accuracy for protein prediction and demonstrated ML's potential for optimizing milk pooling and fortification in human milk banks.

Conclusion: Machine learning enables reliable and faster nutrient estimation, aiding decision-making for milk-bank operations.

Gap/Future Work: Requires larger, more diverse datasets to improve fat prediction accuracy and generalization.

Link: [research paper](#)

Paper 2: Exploring Barriers and Facilitators for Human Milk Donation and Acceptance Among Donors and Recipients

Author: Chaudhari P. & Aiyer S. (2024)

Abstract Summary: The researchers conducted **group discussions** with milk donors and recipients at **SSG Hospital, Vadodara**, identifying motivators.

Conclusion: Social awareness and family involvement are key to improving donation and acceptance rates.

Gap/Future Work: Suggests expanding awareness programs and family-inclusive counselling at multiple centers.

Link: [research paper](#)

2. Prathmesh

Paper 3: Artificial Intelligence Applied to the Study of Human Milk and Breastfeeding: A Scoping Review

Author: Briere C.-E. et al. (2024)

Abstract Summary: Reviewed 19 studies on AI applications in milk and breastfeeding, covering nutrient analysis, breastfeeding prediction, and chatbot-based counselling.

Conclusion: AI can improve efficiency and personalization in milk-bank and breastfeeding support systems.

Gap/Future Work: Data inconsistency and limited sample size hinder large-scale application; future research should focus on integrated AI frameworks.

Link: [research paper](#)

Paper 4: The Application of Machine Learning in the Analysis of Macro and Micro-Nutrients of Human Milk: A Scoping Review

Author: Arzheh A., Seven M., & Briere C.-E. (2025)

Abstract Summary: Analyzed ML approaches for nutrient profiling in milk. Found strong predictive accuracy for macronutrients like protein and fat but limited evidence for micronutrients.

Conclusion: ML is a promising tool for precise nutrient estimation, supporting better milk fortification.

Gap/Future Work: Requires standardized datasets, larger studies, and inclusion of micronutrient prediction models.

Link: [research paper](#)

Paper 5: Application of a Locally Developed Open-Access Digital Monitoring System for the Human Milk Bank Network in Vietnam

Author: Nguyen T. H. et al. (2025)

Abstract Summary: Developed a web-based system (ASP.NET + SQL) for donor and pasteurization tracking, improving data accuracy and scalability.

Conclusion: Digital tools can modernize milk-bank operations with better transparency and reporting.

Gap/Future Work: Expand integration with hospital EMRs and implement nationwide

digital networks.

Link: [research paper](#)

3. Vaibhavi

Paper 6: A Qualitative Analysis of Knowledge, Perception and Practices Related to Human Milk Banking

Author: Jegannathan R. et al. (2023)

Abstract Summary: Interview-based study exploring knowledge gaps among nurses and lactating mothers regarding HMB operations.

Conclusion: Limited awareness and infrastructure impact milk-bank efficiency; continuous training is essential.

Gap/Future Work: Recommend large-scale multi-site training studies to enhance HMB capacity.

Link: [research paper](#)

Paper 7: Awareness and Acceptance of Human Milk Donation

Author: Patel V. et al. (2024)

Abstract Summary: Surveyed 300 mothers; found only 28% awareness but 56% willingness to donate after receiving proper counselling.

Conclusion: Education and communication strategies can significantly improve donation readiness.

Gap/Future Work: Broaden awareness drives in rural and under-informed communities.

Link: [research paper](#)

4. Krushna

Paper 8: A Quality Improvement Project to Improve Voluntary Milk Donation in a Human Milk Bank in South India

Author: Sivanandan S. et al. (2023)

Abstract Summary: Used PDSA cycles and process mapping to increase milk donation through better counselling and workflow. Milk volume rose from 0.83 L to 1.16 L per day.

Conclusion: Simple, low-cost process interventions can enhance donation rates effectively.

Gap/Future Work: Requires multi-center validation and digital donor tracking for sustainability.

Link: [research paper](#)

Paper 9: Sociodemographic and Psychological Factors Influencing Human Milk Donation

Author: Bodi Jayakrishna V., Patil S. V., & Patil M. (2024)

Abstract Summary: Survey of 420 mothers in Vijayapura identified education, awareness, and NICU exposure as major motivators for milk donation.

Conclusion: Knowledge and exposure directly influence willingness to donate.

Gap/Future Work: Long-term tracking and rural-focused awareness programs are needed.

Link: [research paper](#)

Research Gap Identified

1. Limited Dataset Size & Standardization
2. Lack of Integration Between Technology and Practice
3. Insufficient Awareness and Education
4. Single-Center and Short-Term Research
5. Minimal Use of Advanced AI Techniques
6. Weak Digital Infrastructure
7. Lack of Multi-Center Studies

Proposed Future Work

1. Develop **large, multi-center datasets** to improve accuracy and generalization of AI and ML models.
2. **Integrate digital monitoring systems** with hospital EMRs for real-time donor and milk tracking.
3. Launch **nationwide awareness and counselling programs** to increase milk donation and acceptance.
4. Conduct **long-term follow-up studies** to assess the sustained impact of awareness and quality interventions.
5. Combine **AI-driven analysis with social research** to build a holistic, efficient, and inclusive milk-banking ecosystem.